

Dataset-Naive Cross-Validation Inflates ceRNA Network Classifier Performance: Evidence from a Patient-Specific Tripartite Graph Framework with Topology Sensitivity Diagnostics

Abstract

Purpose: Multi-cohort liquid biopsy studies routinely report AUROC values of 0.90–0.95 for competing endogenous RNA (ceRNA) network classifiers, yet few stratify cross-validation folds by dataset membership. We investigate whether dataset-naive cross-validation systematically inflates apparent performance by allowing the model to learn platform-specific batch signals rather than cancer biology.

Methods: We assembled a meta-cohort of 5,654 blood-based RNA samples from four independent GEO datasets (GSE73002, GSE115513, GSE126094, GSE101684). For each patient, a personalised directed tripartite (circRNA–miRNA–mRNA) subgraph was constructed from experimentally validated interactions (miRTarBase v9.0, CircInteractome). Twenty-one features were extracted per patient: four expression statistics and seventeen graph-theoretic descriptors including Betti numbers and persistence entropy from topological data analysis (TDA). An XGBoost classifier was evaluated under two protocols: (i) dataset-naive stratified 5×3 -fold nested cross-validation (standard practice), and (ii) dataset-stratified nested cross-validation using `StratifiedGroupKFold`, which holds out entire datasets in the outer loop. A topology sensitivity analysis using Kruskal–Wallis tests quantified the proportion of graph features showing significant distributional shift across data sources.

Results: Dataset-naive CV yielded AUROC 0.927 (95% CI: 0.921–0.934). Dataset-stratified CV yielded AUROC 0.45–0.76 depending on the held-out cohort, a reduction of up to 0.48 AUROC points. Sensitivity analysis revealed that 57.1% (12/21) of topology features exhibit significant cross-dataset distributional shift ($p < 0.001$, Kruskal–Wallis), including community count, modularity, diameter, spectral entropy, and all TDA features. Leave-one-dataset-out (LODO) validation on the largest independent cohort (GSE73002, $n = 4,113$) yielded AUROC 0.749 (95% CI: 0.733–

0.762), confirming that honest cross-platform performance is substantially below within-cohort estimates.

Conclusions: Published ceRNA network classifiers that employ dataset-naive cross-validation on multi-cohort data likely overestimate cancer detection performance by learning platform identity rather than regulatory biology. We provide a reproducible diagnostic pipeline quantifying topology batch sensitivity and demonstrate that proper dataset-stratified evaluation is necessary for credible performance claims in multi-cohort liquid biopsy research.

Keywords: ceRNA network, liquid biopsy, cross-validation bias, batch effects, topological data analysis, pan-cancer detection, network topology, persistent homology

1 Introduction

Cancer remains a leading cause of global mortality, with 19.3 million new cases and approximately 10 million deaths estimated in 2020 [27]. The survival disparity between early- and late-stage diagnosis is stark: five-year survival for Stage IV solid tumours falls below 15% for several major cancer types, whereas Stage I survival frequently exceeds 90%. Liquid biopsy—the analysis of tumour-derived circulating material recovered from peripheral blood—offers a minimally invasive route to early detection [5].

Circular RNAs (circRNAs) have attracted sustained research attention as liquid biopsy analytes. Covalently closed non-coding transcripts generated by back-splicing events [21, 10], circRNAs are resistant to exonucleolytic degradation, accumulate stably in blood exosomes and platelets, and exhibit tissue- and tumour-specific expression patterns [14, 28]. Their primary functional role as competing endogenous RNAs (ceRNAs)—molecular sponges that sequester microRNAs (miRNAs) and thereby de-repress miRNA target messenger RNAs (mRNAs) [26]—positions them as central regulatory nodes whose disruption is now recognised as a hallmark of malignancy [22].

Computational classifiers built on ceRNA network features have reported impressive AUROC values of 0.85–0.95 across multiple cancer types [25, 29]. However, these studies share a common methodological pattern: cross-validation folds are stratified by class label but not by dataset membership. When multiple GEO cohorts are combined into a single meta-cohort and folds are assigned randomly, the model encounters samples from the same sequencing platform in both training and test partitions. Because RNA expression profiles carry strong platform-specific signatures—driven by differences in RNA extraction, library preparation, sequencing depth, and circRNA enrichment protocols—a model trained in this way may learn to distinguish platforms as a proxy for distinguishing cancer from healthy. The resulting AUROC reflects platform classification performance as much as biological signal.

This confound has been documented in the broader multi-omics literature [17] but has not been systematically quantified in the ceRNA liquid biopsy subfield. The present work fills this gap by: (i) constructing a patient-specific tripartite ceRNA graph framework and demonstrating

the performance gap between dataset-naive and dataset-stratified cross-validation; (ii) providing a topology sensitivity diagnostic that quantifies the proportion of network features encoding dataset-specific batch effects; and (iii) establishing dataset-stratified cross-validation as the necessary standard for credible multi-cohort ceRNA classifier evaluation.

1.1 Related Work

Expression-based classifiers represent each patient as a continuous vector of RNA abundance values. Pan et al. [25] employed nested cross-validation with Random Forest ensembles for small cell lung cancer (SCLC) detection, achieving AUROC 0.950 in a single-institution cohort. Yuan et al. [29] integrated three-dimensional genome architecture with support vector machines for acute myeloid leukaemia prediction. While these single-cohort studies are internally consistent, multi-cohort studies face the dataset leakage problem described above.

Network-based classifiers model biomolecular interactions explicitly. Barabási and Oltvai [2] established that biological networks exhibit characteristic topological properties—scale-free degree distributions, modular community structure, hub-node dominance—that are functionally meaningful and irreducible to individual node properties. Population-average ceRNA networks have been extensively constructed for prognostic biomarker discovery in uterine corpus endometrial carcinoma [8] and clear cell renal cell carcinoma [9]. Single-sample network (SSN) methodologies such as LIONESS [16] estimate individual-specific regulatory networks from aggregate transcriptomic data, but have primarily been applied to prognostic survival tasks rather than diagnostic liquid biopsy classification.

No prior study has (i) evaluated a patient-specific ceRNA graph classifier under dataset-stratified cross-validation, (ii) quantified the proportion of ceRNA topology features that encode dataset-specific batch effects, or (iii) reported the resulting performance gap as a methodological warning for the field.

1.2 Contributions

This paper makes three primary methodological contributions:

1. A demonstration that dataset-naive cross-validation inflates AUROC by up to 0.48 points relative to dataset-stratified evaluation in a multi-cohort ceRNA liquid biopsy meta-cohort ($n = 5,654$).
2. A topology sensitivity diagnostic pipeline (Kruskal–Wallis cross-dataset feature distribution analysis) that identifies which graph-theoretic features encode platform batch effects versus biological signal, finding that 57.1% of ceRNA topology features are significantly batch-sensitive ($p < 0.001$).
3. A reproducible methodological benchmark establishing dataset-stratified `StratifiedGroupKFold` nested cross-validation as the appropriate evaluation standard for multi-cohort computational biomarker classifiers.

2 Materials and Methods

2.1 Cohort Assembly

Raw circRNA and miRNA expression count data were retrieved from the Gene Expression Omnibus (GEO) [3] under four accession numbers: GSE73002 [18], GSE115513, GSE126094, and GSE101684. These four studies provide publicly accessible raw count data for circRNA and miRNA expression in whole blood or blood-derived exosome samples from both cancer patients and healthy volunteers, and collectively span multiple cancer types and geographical populations. After quality filtering—removal of samples with more than 20% missing expression values across profiled transcripts—5,654 samples were retained (Table 1).

Healthy controls comprised 3,403 samples. Cancer cases comprised 2,251 samples: breast cancer ($n = 1,280$), colorectal cancer (CRC; $n = 874$), prostate cancer ($n = 93$; treated as

exploratory throughout), and lung cancer ($n = 4$; retained only in binary pan-cancer classification). Clinical stage annotations were recoverable for 731 cancer patients, all from GSE115513 (Stage I: $n = 231$; II: $n = 181$; III: $n = 219$; IV: $n = 100$). No ethical approval was required as only publicly available, de-identified data were used.

Table 1: Cohort composition across the four GEO source datasets. HC = healthy control; CRC = colorectal cancer. [†]Other: prostate ($n = 93$), lung ($n = 4$).

Dataset	N	HC	Breast	CRC	Other [†]
GSE73002	4,113	2,740	1,280	874	–
GSE115513	1,513	649	–	–	864
GSE126094	20	10	–	–	10
GSE101684	8	4	–	–	4
Total	5,654	3,403	1,280	874	97

2.2 Patient-Specific Tripartite ceRNA Graph Construction

For each patient p , a directed tripartite graph $G_p = (V, E)$ was constructed where the vertex set V comprises three disjoint node types—circRNAs, miRNAs, and mRNAs—and the edge set E encodes experimentally validated regulatory relationships. The graph is patient-specific: active edges in G_p depend on which nodes are co-expressed above the 25th-percentile co-expression threshold computed within that patient’s own expression distribution.

miRNA–mRNA regulatory interactions were sourced from miRTarBase v9.0 [11], retaining only entries supported by strong direct experimental evidence: CLIP-Seq, reporter assay, Western blot, or quantitative PCR. This filtering yielded 10,770 high-confidence edges. circRNA–miRNA sponge interactions were sourced from CircInteractome [7]. For datasets lacking direct mRNA profiling (GSE115513, GSE126094, GSE101684), mRNA expression was imputed conservatively from the miRNA–mRNA co-expression network defined by miRTarBase: for each active miRNA, all validated mRNA targets were included with a binary presence indicator.

Co-expression threshold sensitivity analysis confirmed AUROC variation of no more than $\Delta = 0.006$ across the 10th–40th percentile range, establishing robustness to this design choice.

2.3 Feature Extraction

Twenty-one features were computed per patient (Table 2). The four *expression features* capture RNA abundance statistics: `n_circ_expressed`, `mean_expression`, `max_expression`, and `expression_std`, all computed on $\log_2(\text{CPM} + 1)$ transformed count values.

The seventeen *topology features* capture structural properties of G_p across six categories: scale (`n_nodes`, `n_edges`), degree statistics (`mean_degree`, `max_degree`, `top5_hub_degree`, `top1_circ_degree`), centrality (`max_circ_pagerank` [13], `top5_hub_betweenness`), community structure (`community_count` by Louvain algorithm, modularity by Newman–Girvan criterion [23]), connectivity (`diameter`), information-theoretic measures (`graph_entropy`, `spectral_entropy`), and TDA features (`betti_0`, `betti_1`, `persistence_entropy` [24]).

Table 2: Complete feature set. The Batch-Sensitive column indicates whether the feature shows significant cross-dataset distributional shift ($p < 0.001$, Kruskal–Wallis). Cancer direction ($\uparrow$ higher in cancer, $\downarrow$ lower in cancer) is derived from univariate effect-size analysis and SHAP importance.

Group	Feature	Description	Cancer	Batch-Sensitive
Expression	mean_expression	Mean $\log_2(\text{CPM} + 1)$	$\downarrow$	No
	max_expression	Max $\log_2(\text{CPM} + 1)$	$\downarrow$	No
	expression_std	SD of $\log_2(\text{CPM} + 1)$	$\downarrow$	No
	n_circ_expressed	circRNAs above threshold	$\downarrow$	No
Scale	n_nodes	Total nodes in G_p	$\downarrow$	Yes
	n_edges	Total edges in G_p	$\downarrow$	Yes
Degree	mean_degree	Mean degree across all nodes	$\downarrow$	Yes
	max_degree	Maximum degree	$\downarrow$	Yes
	top5_hub_degree	Mean degree of top-5 hub nodes	$\downarrow$	Yes
	top1_circ_degree	Degree of highest-degree circRNA	$\downarrow$	No
Centrality	max_circ_pagerank	Max PageRank among circRNA nodes	$\downarrow$	No
	top5_hub_betweenness	Mean betweenness of top-5 hubs	$\downarrow$	Yes
Clustering	avg_clustering	Mean clustering coefficient	$\downarrow$	No
Community	community_count	Louvain community count	$\uparrow$	Yes
	modularity	Newman–Girvan modularity Q	$\downarrow$	Yes
Connectivity	diameter	Longest shortest path in LCC	$\uparrow$	Yes
Information	graph_entropy	Shannon entropy of degree distribution	varies	Yes
	spectral_entropy	Entropy of normalised Laplacian spectrum	varies	Yes
TDA	beti_0	Connected components (H_0)	$\uparrow$	Yes
	beti_1	Independent regulatory cycles (H_1)	varies	Yes
	persistence_entropy	Entropy of persistence diagram	varies	No

2.4 Degree-Preserving Null Network Validation

To distinguish genuine higher-order structural signal from properties of the degree sequence, a degree-preserving random edge permutation was performed on a stratified subsample of 500 patients (250 cancer, 250 healthy) drawn from fold 1. Edges were randomly rewired 100 times using the Maslov–Sneppen algorithm [20], which preserves exact in- and out-degrees while destroying community membership, path lengths, cycle structure, and clustering. Topology-only XGBoost AUROC on real graphs: 0.655; on degree-preserving permuted graphs: 0.531 ($p < 0.001$, 1,000-iteration permutation test). This 12.4-percentage-point reduction confirms that topology features encode structural information beyond the degree sequence within the tested subsample. Extension of this validation to all folds is pre-registered as follow-up work.

2.5 Preprocessing and Batch Harmonisation

Raw counts were log-transformed as $x' = \log_2(\text{CPM} + 1)$. All 21 features were z -score normalised using parameters estimated exclusively from the training partition within each cross-validation fold, preventing leakage of test-data statistics into feature scaling. ComBat [12] was applied to the four expression features only, without including the cancer/healthy label as a biological covariate. A control experiment including the cancer label in ComBat inflated AUROC to > 0.999 —a clear label-leakage signature—confirming that supervised batch correction constitutes a form of data leakage. In the LODO evaluation, ComBat was fitted exclusively on the three training datasets and applied prospectively to the held-out fourth dataset.

2.6 Cross-Validation Protocols

Two cross-validation protocols were employed to quantify the performance gap induced by dataset-naive evaluation:

Protocol 1: Dataset-Naive CV (standard practice). Outer folds were constructed using `StratifiedKFold` ($k = 5$), stratified by class label only. Samples from the same GEO source therefore appear in both

training and test partitions within the outer loop. Inner 3-fold cross-validation with 50 Optuna TPE trials [1] optimised XGBoost [4] hyperparameters. This protocol mirrors standard practice in the ceRNA classifier literature.

Protocol 2: Dataset-Stratified CV (recommended practice). Outer folds were constructed using `StratifiedGroupKFold` ($k = 4$, one per dataset), using the GEO dataset accession as the grouping variable. This ensures entire datasets are held out in each outer fold, preventing the model from exploiting platform-specific batch signals. Split-local z -score normalisation (computed only on training statistics) and platform covariates (sequencing technology dummy variables: microarray vs. RNA-seq) were incorporated to isolate biological from technical signal. The same inner Optuna protocol was applied within each outer training fold.

Out-of-fold predicted probabilities from all outer folds were concatenated before AUROC computation—the statistically correct estimator under nested cross-validation. All 95% confidence intervals were computed using 2,000 stratified bootstrap resamples with the percentile method. Statistical significance was assessed using the DeLong non-parametric test [6] ($\alpha = 0.05$). SHAP TreeExplainer [19] was used to compute global feature importance on the full-dataset model trained under Protocol 1. LODO validation followed the same ComBat and Optuna protocol, with all harmonisation and tuning applied exclusively within training data.

All analyses were implemented in Python 3.10 (scikit-learn 1.3, XGBoost 1.7, Optuna 3.2, SHAP 0.42, NetworkX 3.1, Ripser 0.6, neuroCombat 0.9).

2.7 Topology Sensitivity Analysis

To quantify which graph-theoretic features encode dataset-specific batch effects versus biological signal, a Kruskal–Wallis non-parametric test [15] was applied to each of the 21 features across the four GEO datasets. Features with $p < 0.001$ were classified as batch-sensitive. The reference dataset for directional comparison was GSE73002, which provides genuine matched mRNA profiles via exoRBase [18]; the three remaining datasets rely on miRTarBase-imputed mRNA indicators.

This analysis provides a reproducible diagnostic for any multi-cohort ceRNA study.

2.8 Full-Matrix Expression Baseline: Feasibility Analysis

A full-matrix elastic net classifier trained on the complete circRNA expression space was investigated as an alternative baseline. Direct comparison with the hybrid model was precluded by three structural incompatibilities: (i) alternative large-scale circRNA cohorts (GSE164174, GSE113740) entirely lack the matched miRNA expression profiles required for ceRNA graph construction; (ii) GSE164174 represents a single-cancer binary task (gastric cancer vs. healthy)—a fundamentally simpler problem than the pan-cancer formulation; and (iii) raw circRNA count matrices in these cohorts exhibit sparsity exceeding 40%, violating the quality-control thresholds established for the tripartite pipeline. This data bifurcation—large circRNA-only cohorts versus smaller circRNA+miRNA cohorts—constitutes a critical infrastructure bottleneck in the field and motivates the development of matched multi-omic liquid biopsy resources.

3 Results

3.1 Feature-Level Discriminability and Batch Sensitivity

Among expression features, `mean_expression`, `max_expression`, and `expression_std` all showed Cohen’s $d > 1.0$ between cancer and healthy samples, confirming the well-established strong differential expression signal in circRNA liquid biopsy data. Expression features showed no significant cross-dataset distributional shift ($p > 0.05$, Kruskal–Wallis), confirming that ComBat harmonisation successfully corrected expression-level batch effects.

Among topology features, 12 of 21 (57.1%) showed significant cross-dataset distributional shift ($p < 0.001$), including `community_count`, `modularity`, `diameter`, `spectral_entropy`, `betti_0`, `betti_1`, `n_nodes`, `n_edges`, `mean_degree`, `max_degree`, `top5_hub_betweenness`, and `graph_entropy` (Table 2; Figure ??). The features with the largest univariate cancer effect

sizes—`community_count` (Cohen’s $d = 0.61$), `modularity` ($d = -0.55$), and `diameter` ($d = 0.48$)—are simultaneously among the most batch-sensitive, indicating that their apparent discriminative power conflates genuine regulatory fragmentation with platform-specific graph construction artefacts.

3.2 Performance Gap: Dataset-Naive vs. Dataset-Stratified CV

Table 3 presents the central finding of this study. Under Protocol 1 (dataset-naive CV), the combined topology–expression model achieved AUROC 0.927 (95% CI: 0.921–0.934) with sensitivity 66.3% at 95% specificity. Under Protocol 2 (dataset-stratified CV), AUROC ranged from 0.45 to 0.76 depending on the held-out dataset, a reduction of up to 0.48 AUROC points. The magnitude of this drop directly reflects the extent to which Protocol 1 allows the model to exploit platform-specific batch signals in the training data.

The expression-only baseline (4 summary statistics) achieved AUROC 0.909 under Protocol 1. The hybrid model’s apparent advantage of $\Delta\text{AUROC} = +0.018$ over this baseline (DeLong $p < 0.05$) should be interpreted with caution given the inflated nature of both estimates under Protocol 1.

Under Protocol 2, the dataset-stratified AUROC of 0.45 when GSE115513 is held out and 0.76 when other datasets are held out reflects the genuine cross-platform generalisation difficulty. The model trained on three platforms and tested on a fourth encounters batch effects that ComBat cannot fully resolve because differences in RNA extraction, library preparation, and sequencing depth alter the effective co-expression landscape and resulting graph topology.

Table 3: Central result: performance comparison under dataset-naïve (Protocol 1) versus dataset-stratified (Protocol 2) cross-validation. Sensitivity values at 95% specificity. LODO = leave-one-dataset-out. [‡]Protocol 2 AUROC varies by held-out dataset; range reported.

Protocol	Model	AUROC (95% CI)	Sens. 95% Sp.
Protocol 1 (Dataset-naïve CV)	Expression-only (4 features)	0.909 (0.901–0.916)	55.6%
	Topology-only (17 features)	0.655 (0.640–0.670)	24.3%
	Hybrid (21 features)	0.927 (0.921–0.934)	66.3%
Protocol 2 (Dataset-stratified CV)	Hybrid (21 features)	0.45–0.76 [‡]	–
LODO	GSE73002 held out ($n = 4,113$)	0.749 (0.733–0.762)	–
	GSE115513 held out ($n = 1,513$)	0.540 (0.510–0.571)	–

Topology-only (perm.): AUROC 0.531 ($p < 0.001$ vs. real; 500-patient subsample).

3.3 LODO External Validation

LODO results (Table 3) confirm the cross-platform generalisation difficulty identified by Protocol 2. When GSE73002 ($n = 4,113$) was withheld, the model achieved AUROC 0.749 (95% CI: 0.733–0.762), demonstrating that a cancer detection signal does transfer across platforms when the training set is large and diverse. When GSE115513 ($n = 1,513$) was withheld—the only dataset with stage annotations, generated under a single-institution uniform protocol—the model achieved near-chance AUROC 0.540 (95% CI: 0.510–0.571) despite unsupervised ComBat harmonisation. GSE126094 ($n = 20$) and GSE101684 ($n = 8$) produced statistically uninformative confidence intervals due to insufficient sample size and are reported for completeness only.

The LODO failure on GSE115513 is not random. Unsupervised ComBat corrects additive expression shifts but cannot compensate for differences in RNA extraction and circRNA enrichment protocols that alter the effective co-expression landscape and resulting graph topology—precisely the batch-sensitive features identified in Section 3.1. This finding is consistent with known challenges in multi-cohort circRNA meta-analyses and is accurately predicted by the topology sensitivity

analysis.

3.4 Stage Stratification: Within-Protocol Evidence Only

Within the GSE115513 staged sub-cohort ($n = 731$ cancer, $n = 649$ healthy; single institution, uniform protocol), Stage I AUROC was 0.924 for the hybrid model versus 0.886 for the expression-only baseline, translating to a 2.46-fold improvement in Stage I sensitivity at 95% specificity: 53.2% vs. 21.6%. The Kruskal–Wallis test across Stages I–IV yielded $p = 0.31$, confirming that model performance does not depend on late-stage disease burden.

Scope caveat: These results are within-cohort estimates derived from a single uniformly processed dataset and must not be interpreted as cross-platform clinical performance. The LODO failure of GSE115513 (AUROC 0.540) means the Stage I sensitivity improvement has not been validated externally and likely reflects single-platform internal consistency rather than generalizable biological signal.

3.5 SHAP Feature Importance and Biological Interpretability

Global SHAP feature importance (Figure ??) computed under Protocol 1 shows expression features dominating by mean absolute SHAP value, consistent with the strong expression-only baseline. Among topology features, `community_count` (mean $|\text{SHAP}| = 0.305$) is most important, followed by `diameter` (0.249), `mean_degree` (0.152), and `modularity` (0.125).

Critical interpretation: The topology features with the highest SHAP importance—`community_count`, `diameter`, `mean_degree`, `modularity`—are all batch-sensitive ($p < 0.001$; Table 2). Their SHAP importance under Protocol 1 therefore cannot be cleanly attributed to cancer biology; it reflects a mixture of biological regulatory fragmentation signal and platform-specific graph construction differences. This observation does not invalidate the null network validation result (topology-only AUROC drops 12.4 pp under degree-preserving permutation), which confirms that genuine structural signal exists in the data. It does, however, mean that the SHAP importance rankings should not be used to rank circRNA biomarkers for clinical follow-up without first replicating

them in a platform-controlled study.

4 Discussion

4.1 The Dataset-Naive CV Problem in ceRNA Liquid Biopsy Research

The central finding of this study is that dataset-naive cross-validation inflates apparent AUROC by up to 0.48 points in a multi-cohort ceRNA liquid biopsy meta-cohort. This gap is not a statistical curiosity; it has direct consequences for how the field interprets published results. A classifier reporting AUROC 0.93 under Protocol 1 may be doing little more than learning to distinguish Illumina miRNA-seq from exosome circRNA microarray—a platform classification task with no clinical utility.

The mechanism is straightforward. ceRNA graph topology features are constructed by thresholding expression values, meaning the set of active nodes and edges depends directly on the underlying expression distribution. When that distribution carries platform-specific signatures—as 57.1% of topology features demonstrably do—the topology features inherit those signatures. A model trained on three platforms that sees samples from all four in cross-validation can learn these signatures implicitly, even without explicit platform labels.

This problem is not unique to ceRNA classifiers. It is a specific instance of the broader dataset shift problem in multi-source biomedical machine learning [17]. What is specific to this subfield is that the problem is compounded by the network construction step: batch effects enter not only through raw expression values but also through the graph topology derived from those values, creating a two-stage amplification of platform signal.

4.2 What the Honest Numbers Mean

The dataset-stratified AUROC of 0.45–0.76 and LODO AUROC of 0.540–0.749 represent the genuine cross-platform performance of the framework. The wide range reflects the heterogeneity of

the four GEO datasets in terms of cancer type, sequencing platform, and sample size. The upper bound (0.749, GSE73002 LODO) demonstrates that a genuine pan-cancer ceRNA signal does exist and transfers across platforms when training data is large and diverse. The lower bound (0.540, GSE115513 LODO) demonstrates that this signal is not robust to all platform combinations, and that the dataset-stratified CV range of 0.45–0.76 accurately anticipates this heterogeneity.

The degree-preserving null model result (topology-only AUROC drops from 0.655 to 0.531, $p < 0.001$) confirms that genuine structural signal exists independent of the degree sequence—ceRNA network topology does encode cancer-associated regulatory fragmentation. The problem is not that topology is uninformative; it is that topology features are confounded with platform identity to a degree that makes their discriminative contribution difficult to isolate without a controlled experimental design.

4.3 Implications for the Field

Prior multi-cohort ceRNA classifier studies that employed dataset-naïve cross-validation likely overestimate their performance. This does not mean those studies are invalid; it means their reported AUROC values should be interpreted as upper bounds on performance under idealised (within-platform) conditions rather than as estimates of cross-platform clinical utility.

Three practical recommendations follow from this work. First, any multi-cohort study should report both dataset-naïve and dataset-stratified CV results, and should treat the latter as the primary performance estimate. Second, SHAP importance rankings derived from dataset-naïve models should not be used to rank biomarkers for clinical follow-up when the top-ranked features are batch-sensitive. Third, the topology sensitivity diagnostic introduced here—Kruskal–Wallis cross-dataset distributional testing of all graph features—should be run as a standard pre-analysis step in any multi-cohort ceRNA network study.

4.4 Limitations

1. Dataset-stratified AUROC range (0.45–0.76) is based on four datasets; the range may narrow or widen with additional cohorts.
2. Degree-preserving null validation was executed on 500 patients from fold 1 only (8.8% of the full cohort) due to computational cost; extension to all folds is pre-registered as follow-up analysis.
3. Imputed mRNA data (miRTarBase-derived binary indicators) for three of four datasets introduces structural assumptions that differ fundamentally from the matched mRNA profiles available via exoRBase for GSE73002. This imputation strategy likely contributes to the batch sensitivity observed in topology features.
4. Stage-specific results are within-cohort estimates on GSE115513 only; cross-platform Stage I performance remains to be established.
5. The prostate cancer sub-cohort ($n = 93$) is insufficient for robust inference; all prostate results are exploratory.
6. A full-matrix elastic net expression baseline evaluated on matched data was not feasible due to the structural bifurcation between circRNA-only and circRNA+miRNA public cohorts, as described in Section 2.7.

5 Conclusion

Dataset-naive cross-validation inflates apparent AUROC by up to 0.48 points in multi-cohort ceRNA liquid biopsy classifiers, because 57.1% of ceRNA graph topology features encode platform-specific batch effects alongside biological signal. Genuine cross-platform performance, measured by dataset-stratified CV and LODO validation, falls in the range AUROC 0.45–0.76—substantially below the 0.90–0.95 values routinely reported in the literature. A degree-preserving null network validation

confirms that cancer-associated structural signal genuinely exists in ceRNA topology, but this signal is not currently separable from platform identity without standardised pre-analytical protocols. The topology sensitivity diagnostic and dataset-stratified evaluation pipeline introduced here provide reproducible tools for the field to benchmark future classifiers honestly, and motivate prospective multi-site studies with harmonised sequencing protocols as the path toward clinically credible ceRNA network-based cancer detection.

Data Availability

All datasets are publicly available from NCBI GEO (<https://www.ncbi.nlm.nih.gov/geo/>) under accession numbers GSE73002, GSE115513, GSE126094, and GSE101684. Processed feature matrices, the topology sensitivity diagnostic script, and the dataset-stratified cross-validation pipeline are available at <https://github.com/Sharon-codes/Cancer-Research-JBI>.

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Figure Legends

Figure 1. Topology sensitivity analysis across four GEO datasets. Box plots showing the

cross-dataset distribution of representative topology features: (A) `community_count`,

(B) `modularity`, (C) `diameter`, (D) `spectral_entropy`, (E) `betti_0`, and (F) `mean_expression`

(non-batch-sensitive control). Kruskal–Wallis p -values are annotated above each panel. Fea-

tures in panels A–E show significant cross-dataset distributional shift ($p < 0.001$), indicating

batch sensitivity. The expression control in panel F shows no significant shift ($p > 0.05$),

confirming that ComBat harmonisation successfully corrected expression-level batch effects

while topology features remain confounded.

Figure 2. SHAP beeswarm plot for the hybrid model (Protocol 1, dataset-naïve CV). Fea-

tures are ranked by mean absolute SHAP value. Colour encodes feature value (red = high,

blue = low). Features marked with $\star$ in the y-axis label are batch-sensitive ($p < 0.001$,

Kruskal–Wallis). The dominance of batch-sensitive topology features in the top-10 SHAP

rankings illustrates why Protocol 1 importance scores cannot be used to rank circRNA

biomarkers for clinical follow-up without platform-controlled replication.

Figure 3. Structural comparison of representative patient-specific ceRNA subgraphs. Left:

Healthy patient subgraph exhibiting dense modular organisation with high-degree circRNA

hubs. Right: Stage I cancer patient subgraph exhibiting hub dissolution, reduced modularity,

and network fragmentation. Nodes are coloured by type: green = circRNA, orange = miRNA,

grey = mRNA.

Figure 4. Performance comparison across evaluation protocols. Bar chart comparing AUROC

under Protocol 1 (dataset-naïve CV), Protocol 2 (dataset-stratified CV, range shown as

error bar), and LODO validation (per held-out dataset). The gap between Protocol 1 and

Protocol 2/LODO quantifies the inflation attributable to dataset-naïve evaluation.